

AZURE ENTERPRISE INFRASTRUCTURE LAB

05 — Azure SQL / SQL Server on Azure VM

Complete Portfolio Documentation

1. Project Scope

This portfolio entry documents two related Microsoft SQL deployment patterns used in the Azure Enterprise Lab: Azure SQL Database (PaaS) and the planned installation of Microsoft SQL Server directly inside the existing Windows VM vm-web-MM (IaaS).

Important status note: The attached evidence proves the Azure SQL deployment, VM/RDP connectivity, and supporting Azure configuration. The actual SQL Server installer/service/SSMS installation inside the VM has not yet been captured in this conversation, so that portion is documented as a planned implementation.

2. Architecture

- Azure Subscription → rg-azure-enterprise_MM
- rg-azure-enterprise_MM → vm-web-MM → Windows Server 2025
- vm-web-MM → planned SQL Server 2022 Developer Edition
- rg-azure-enterprise_MM → sqlserver-enterprise-mm-2026 → sqlldb-enterprise-MM

3. Existing Azure VM

Property	Value
VM	vm-web-MM
OS	Windows Server 2025 Datacenter
Size	Standard B2ts_v2
Region	Central India
Availability Zone	2
VNet	vnet-azure-enterprise-MM
Subnet	WebSubnet_MM
Private IP	10.0.1.4
Public IP	20.244.97.131
NIC	vm-web-mm993

4. Existing Azure SQL Database

Property	Value
Database	sqlldb-enterprise-MM
Logical Server	sqlserver-enterprise-mm-2026
Server FQDN	sqlserver-enterprise-mm-2026.database.windows.net
Region	Central India
Port	TCP 1433
Free offer	Applied during creation flow

5. SQL Server on VM — Planned Implementation

1. RDP into vm-web-MM.
2. Download Microsoft SQL Server Developer Edition inside the VM.
3. Start SQL Server Installation Center.
4. Choose New SQL Server stand-alone installation.
5. Select Developer edition.
6. Select Database Engine Services.
7. Use the Default instance / MSSQLSERVER.
8. Use installer-default service configuration for the learning lab.
9. Select Mixed Mode authentication.
10. Set a strong SQL administrative password and add the current Windows administrator.
11. Install and verify completion.

Never store SQL credentials in GitHub or screenshots.

6. Verify SQL Server Service

Run in the Windows VM:

```
Get-Service MSSQLSERVER
```

Expected state: Status = Running, Name = MSSQLSERVER

7. Install SSMS and Connect

- Install SQL Server Management Studio separately.
- Server type: Database Engine.
- Server name: localhost.
- Authentication: Windows Authentication.
- Connect and verify the Databases node.

8. Create Local SQL Database

Run in SSMS:

```
CREATE DATABASE EnterpriseLabDB;
GO
USE EnterpriseLabDB;
GO
CREATE TABLE Customers
(
    CustomerID INT PRIMARY KEY,
    CustomerName VARCHAR(100),
    City VARCHAR(100)
);
GO
INSERT INTO Customers (CustomerID, CustomerName, City)
```

VALUES

```
(1, 'Customer One', 'Kolkata'),  
(2, 'Customer Two', 'Delhi'),  
(3, 'Customer Three', 'Mumbai');  
GO  
SELECT * FROM Customers;  
GO
```

9. VM-to-Azure SQL Connectivity

The VM outbound public IP was verified from PowerShell as 20.244.97.131. For a basic connectivity test, the Azure SQL logical server firewall can allow this specific client address.

Test command:

```
Test-NetConnection sqlserver-enterprise-mm-2026.database.windows.net -Port 1433
```

Expected result: TcpTestSucceeded : True

10. Azure SQL Firewall

Field	Recommended lab value
Rule name	Allow-VM-Web-MM
Start IP	20.244.97.131
End IP	20.244.97.131
Port	1433
Protocol	TCP

Avoid broad 0.0.0.0/0-style exposure for the portfolio lab.

11. Cost Control

The SQL creation workflow initially displayed an expensive Hyperscale estimate. The Free offer was then applied and the portal showed zero displayed compute and storage cost with overage billing disabled. This demonstrates cost-aware deployment selection.

12. Security Considerations

- Patch Windows Server and SQL Server.
- Restrict inbound TCP 1433.
- Avoid exposing SQL Server directly to the public internet unless required.
- Prefer private networking for production-style SQL deployments.
- Use least-privilege access.
- Never publish passwords or connection secrets.

13. Interview Skills

- PaaS vs IaaS for SQL
- Windows Server administration

- SQL Server installation
- Database Engine configuration
- Mixed Mode authentication
- SQL Server service management
- SSMS
- T-SQL
- TCP 1433
- Azure SQL firewall
- Cost management
- RDP troubleshooting

14. Interview Questions

Q1. Azure SQL Database vs SQL Server on Azure VM?

Azure SQL Database is PaaS; Microsoft manages most underlying infrastructure. SQL Server on a VM is IaaS; you manage the operating system and SQL Server.

Q2. Why install SQL Server on the VM?

To gain full SQL Server administration experience, including instance configuration, services, patching, backups and OS/SQL integration.

Q3. What is the default SQL Server instance?

MSSQLSERVER.

Q4. What port does SQL Server commonly use?

TCP 1433 for the default Database Engine instance.

Q5. Why use Mixed Mode?

It supports Windows authentication and SQL Server authentication.

Q6. Why not expose 1433 to Any?

It increases the attack surface; restrict to trusted sources or use private networking.

Q7. Why use Developer Edition?

For development/testing and learning; it is not a production licensing choice.

15. Evidence Collected

The current documentation package includes 13 attached screenshots covering the Azure SQL lifecycle, VM connectivity and Azure configuration. The actual SQL Server-on-VM installer/service/SSMS screenshots remain to be captured.

16. Required SQL Server-on-VM Screenshots

- SQL Server installer landing page
- Feature Selection
- Instance Configuration
- Mixed Mode Authentication

- Installation success
- SQL Server (MSSQLSERVER) service running
- SSMS connected to localhost
- EnterpriseLabDB created
- T-SQL result

17. GitHub Structure

- 05-azure-sql/README.md
- 05-azure-sql/screenshots/
- 05-azure-sql/sql-server-on-vm/screenshots/
- 05-azure-sql/azure-sql-database/

18. Git Commit

```
cd /d C:\azure-enterprise-infrastructure-lab
git add 05-azure-sql
git commit -m "Document Azure SQL and SQL Server on VM lab"
git push
```

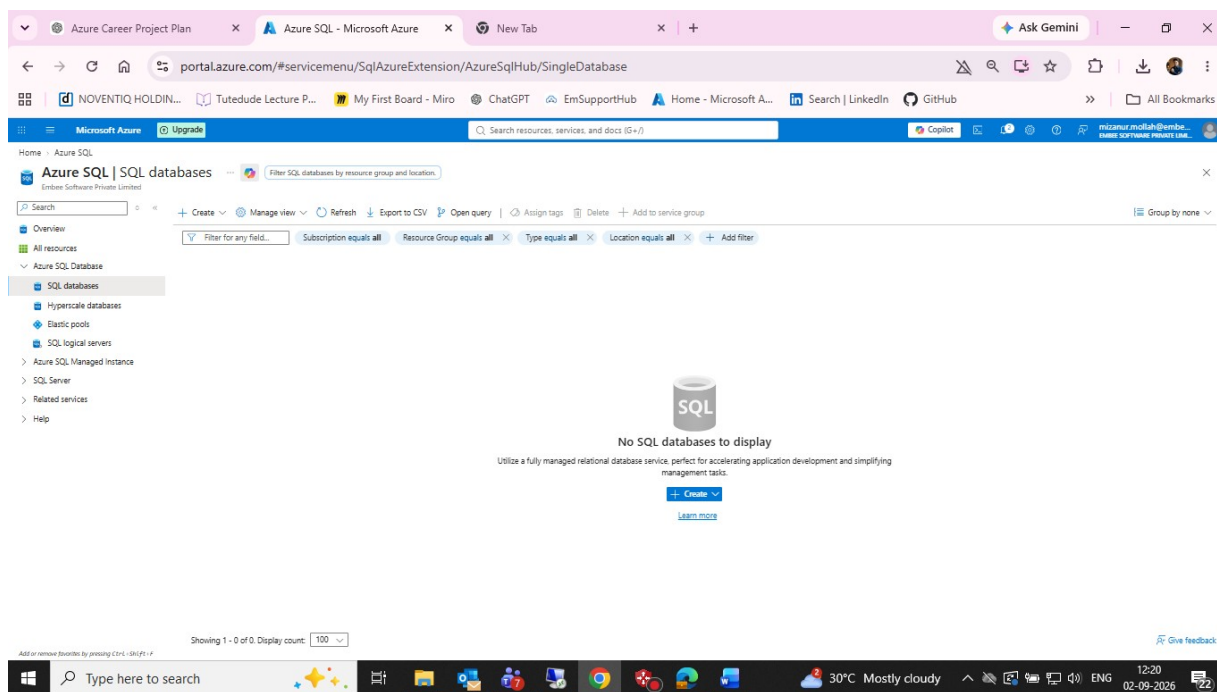
19. Completion Checklist

- ☒ Azure SQL Database deployed
- ☒ SQL logical server deployed
- ☒ Free SQL offer applied
- ☒ Cost checked before deployment
- ☒ VM deployed and running
- ☒ RDP verified
- ☒ VM outbound IP verified
- ☒ Azure SQL Query Editor accessed
- ☒ Resource-group evidence captured
- ☐ Install SQL Server Developer Edition on VM
- ☐ Configure Database Engine
- ☐ Configure default instance
- ☐ Configure Mixed Mode
- ☐ Add Windows administrator
- ☐ Verify MSSQLSERVER service

- ☐ Install SSMS
- ☐ Connect to localhost
- ☐ Create EnterpriseLabDB
- ☐ Execute T-SQL
- ☐ Capture SQL Server installer screenshots
- ☐ Review SQL port 1433 security
- ☐ Commit and push

20. Attached Evidence Screenshots

Figure 1 — Azure SQL databases resource page before creation



Azure SQL databases resource page before creation

Figure 2 — Create SQL Database Server form

The screenshot shows the 'Create SQL Database Server' form in the Azure portal. The browser tabs include 'Azure Career Project Plan', 'Create SQL Database Server - N...', and 'New Tab'. The address bar shows the URL: `portal.azure.com/#view/SqIAzureExtension/SharedCreateServer/isNewServer~/true/subscriptionId/14ecae1d-b340-466e-82a5-3ae59a77...`. The Microsoft Azure header is visible with a search bar and 'Upgrade' button. The breadcrumb trail is 'Home > Azure SQL | SQL databases > Create SQL Database'. The page title is 'Create SQL Database Server'. Below the title, there's a 'Server details' section with instructions: 'Enter required settings for this server, including providing a name and location. This server will be created in the same subscription and resource group as your database.' The 'Server name' field has a placeholder 'Enter server name' and a dropdown showing '.database.windows.net'. The 'Location' field has a placeholder 'Loading...'. The 'Authentication' section has a blue banner stating 'Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)'. Below this, it says 'Select your preferred authentication methods for accessing this server. We recommend using only Microsoft Entra authentication. [Learn more](#) or using an existing Microsoft Entra user, group, or application as Microsoft Entra admin. [Learn more](#).' The 'Authentication method' section has three radio buttons: 'Use Microsoft Entra-only authentication' (selected), 'Use both SQL and Microsoft Entra authentication', and 'Use SQL authentication'. The 'Set Microsoft Entra admin' section shows the email 'mizanur.molish@emboss.co.in', the role 'Admin Object/App ID: 86a5719f-9942-41d8-b879-c770702b7876', and a 'Set admin' link. At the bottom, there's an 'OK' button and a 'Feedback' link. The Windows taskbar at the very bottom shows the search bar, taskbar icons, and system tray with weather '30°C Mostly cloudy' and date '12:23 02-09-2026'.

Server details

Enter required settings for this server, including providing a name and location. This server will be created in the same subscription and resource group as your database.

Server name *

Enter server name .database.windows.net

Location

Loading...

Authentication

Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. We recommend using only Microsoft Entra authentication. [Learn more](#) or using an existing Microsoft Entra user, group, or application as Microsoft Entra admin. [Learn more](#).

Authentication method

☒ Use Microsoft Entra-only authentication

☐ Use both SQL and Microsoft Entra authentication

☐ Use SQL authentication

Set Microsoft Entra admin

mizanur.molish@emboss.co.in

Admin Object/App ID: 86a5719f-9942-41d8-b879-c770702b7876

[Set admin](#)

OK Feedback

Create SQL Database Server form

Figure 3 — SQL logical server authentication configured

The screenshot shows the 'Create SQL Database Server' page in the Azure portal. The page is titled 'Create SQL Database Server' and includes a sub-header 'Microsoft'. Below the header, there is a section for 'Enter required settings for this server, including providing a name and location. This server will be created in the same subscription and resource group as your database.' The settings are as follows:

- Server name ***: sqlserver-enterprise-mm-2026 (with a checkmark icon)
- Location ***: (Asia Pacific) Central India (with a checkmark icon)
- Authentication**: A section with a blue header and a link to 'Learn more'.
- Select your preferred authentication methods for accessing this server. We recommend using only Microsoft Entra authentication. Learn more or using an existing Microsoft Entra user, group, or application as Microsoft Entra admin. Learn more or**
- Authentication method**: Three radio buttons are present: 'Use Microsoft Entra-only authentication' (unselected), 'Use both SQL and Microsoft Entra authentication' (selected), and 'Use SQL authentication' (unselected).
- Set Microsoft Entra admin**: A text field containing 'mizansur.mollah@embee.co.in' and a link to 'Set admin'.
- Server admin login ***: sqladminmm (with a checkmark icon)
- Password ***: A masked password field (with a checkmark icon)
- Confirm password ***: A masked password field (with a checkmark icon)

At the bottom of the page, there is a blue 'OK' button and a 'Feedback' link. The Windows taskbar is visible at the bottom of the screen, showing the search bar, task view button, and various application icons. The system tray shows the date and time as 12:25 on 02-09-2026.

SQL logical server authentication configured

Figure 4 — SQL deployment started

The screenshot displays the Azure portal interface for a deployment named 'CreateSqlDatabase-1788332175099'. The main section shows the deployment is 'in progress'. A table lists the deployment details:

Resource	Type	Status	Operation details
sqlserver-enterprise-mm-2026	Microsoft.Sql/servers	Accepted	Operation details

On the right, a 'Notifications' pane shows a list of events:

- Deployment in progress... Running (a few seconds ago)
- Successfully uploaded blob(s) (9 minutes ago)
- Successfully created storage container (10 minutes ago)
- Deployment succeeded (11 minutes ago)
- Started virtual machine (22 minutes ago)
- ¥19,065.30 credit remaining (22 minutes ago)

SQL deployment started

Figure 5 — SQL server operation details while running

The screenshot displays the Azure portal interface for a deployment. The main window shows the 'CreateSqlDatabase-1788332175099' deployment in progress. The deployment details table shows a resource of 'sqlserver-enterprise-mm-2026' with a status of 'Accepted'.

Resource	Type	Status
sqlserver-enterprise-mm-2026	Microsoft.Sql/servers	Accepted

The 'Operation details' pane on the right provides a comprehensive log of the deployment process, including the deployment correlation ID, operation ID, operation name, provisioning operation, provisioning state, timestamp, duration, tracking ID, service request ID, status, type, resource ID, and resource name.

Operation details:

- Deployment correlation ID: 82d4c583-527e-4628-9d9d-ec41167ac3e3
- Operation ID: /subscriptions/14ecae1d-b340-466e-82a5-3ae59a77c51b/resourceGroups/rg-azur.../providers/Microsoft.Sql/servers/sqlserver-enterprise-mm-2026/operations/82d4c583-527e-4628-9d9d-ec41167ac3e3
- Operation name: Create
- Provisioning operation: Create
- Provisioning state: Running
- Timestamp: 9/2/2026, 12:26:24 PM
- Duration: 27 seconds
- Tracking ID: 13e10bc7-d890-4033-8f9f-93b02c09a838
- Service request ID: 8c8c5a8-047d-4358-bd5d-d56e9e25c1bd
- Status: Accepted
- Type: Microsoft.Sql/servers
- Resource ID: /subscriptions/14ecae1d-b340-466e-82a5-3ae59a77c51b/resourceGroups/rg-azur.../providers/Microsoft.Sql/servers/sqlserver-enterprise-mm-2026
- Resource: sqlserver-enterprise-mm-2026

The 'Notifications' pane on the right shows a list of events in the activity log, including 'Deployment in progress', 'Successfully uploaded blob(s)', 'Successfully created storage container', 'Deployment succeeded', and 'Started virtual machine'.

SQL server operation details while running

Figure 6 — Resource group showing SQL database and SQL server

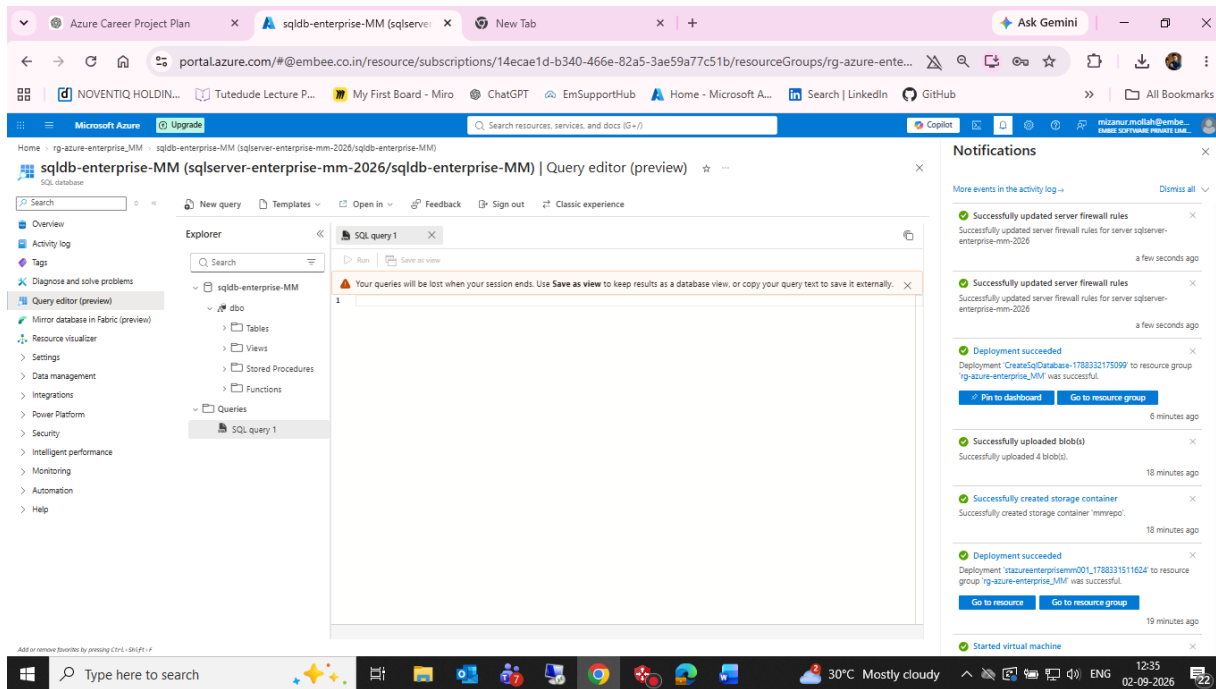
The screenshot shows the Azure portal interface for the resource group 'rg-azure-enterprise_MM'. The 'Resources' tab is active, displaying a list of resources. The resources are filtered by 'Type equals all' and 'Location equals all'. The list includes:

Name	Type	Location
ns-g-vm-MM	Network security group	Central India
sqldb-enterprise-MM (sqlserver-enterprise-mm-2026/sqldb-enterprise-MM)	SQL database	Central India
sqlserver-enterprise-mm-2026	SQL server	Central India
storageenterprise-mm001	Storage account	Central India
vm-web-MM	Virtual machine	Central India
vm-web-MM-ip	Public IP address	Central India
vm-web-MM-nsg	Network security group	Central India
vm-web-mm693	Network interface	Central India
vm-web-MM_OsDisk_1_0c85850c016466fa2b0cc4e7ab644a2	Disk	Central India
vnet-azure-enterprise-MM	Virtual network	Central India
vnet-azure-enterprise-MM-bastion	Bastion	Central India

The SQL database and SQL server resources are highlighted in the list. The bottom of the screen shows the Windows taskbar with the search bar, task icons, and system tray information including the date and time (12:45, 02-09-2026).

Resource group showing SQL database and SQL server

Figure 7 — Azure SQL Query Editor



Azure SQL Query Editor

Figure 8 — SQL database creation screen showing an expensive Hyperscale estimate

The screenshot shows the 'Create SQL Database' wizard in the Azure portal. The 'Basics' tab is selected, showing configuration options for a serverless Azure SQL Database. The 'Project details' section shows the subscription 'My Test Azure subscription 1' and resource group 'rg-azure-enterprise_XM'. The 'Database details' section shows the database name 'Enter database name' and server 'sqlserver-enterprise-mm-2026 (Central India)'. The 'Cost summary' panel on the right shows a high estimate for the Hyperscale configuration.

Cost summary

Configuration	Cost per vCore (in USD)	Quantity	Total Cost
Hyperscale (M5, Gen5, 2) - Primary replica	146.68	x 2	293.36
vCores selected			
Hyperscale (M5, Gen5, 2) - HA replica	146.68	x 2	293.36
vCores selected			
HA replicas		x 1	
ESTIMATED COMPUTE COST / MONTH	\$86.70 USD		
STORAGE COST / GB / MONTH	0.28 USD		

STORAGE NOTES
* In the hyperscale tier, storage costs are calculated based on actual allocation. Allocated space increases automatically as needed, up to 100 TB.

SQL database creation screen showing an expensive Hyperscale estimate

Figure 9 — SQL Free offer applied with zero displayed compute/storage cost

The screenshot shows the 'Create SQL Database' wizard in the Azure portal. The 'Basics' tab is active, displaying the 'SQL Free offer' which provides 100,000 vCore seconds, 32GB of data, and 32GB of backup storage free per month. The 'Cost summary' panel on the right shows a total cost of 0.00 USD, with a note that the first 32GB of storage is free. The 'Database details' section shows the database name 'Enter database name' and the server 'sqlserver-enterprise-mm-2026 (Central India)'. The 'Next: Networking' button is visible at the bottom.

Home

Create SQL Database

Basics Networking Security Additional settings Tags Review + create

Create a SQL database with your preferred configurations. Complete the Basics tab then go to Review + Create to provision with smart defaults, or visit each tab to customize. [Learn more](#)

Free offer applied! You get 100,000 vCore seconds, 32GB of data, and 32GB of backup storage free per month for the lifetime of your subscription. [Learn more](#)

Remove offer

SQL Database Hyperscale: Low price, high scalability, and best feature set. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group * [Create new](#)

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources.

Database name *

Server *

[Review + create](#) [Next: Networking >](#)

Cost summary

General Purpose (GP-S, Gen5-2) 0.00

Cost per GB (in USD) x 416

Max storage selected (in GB)

First 32 GB storage free

First 100,000 vCore seconds free

Overage billing Disabled

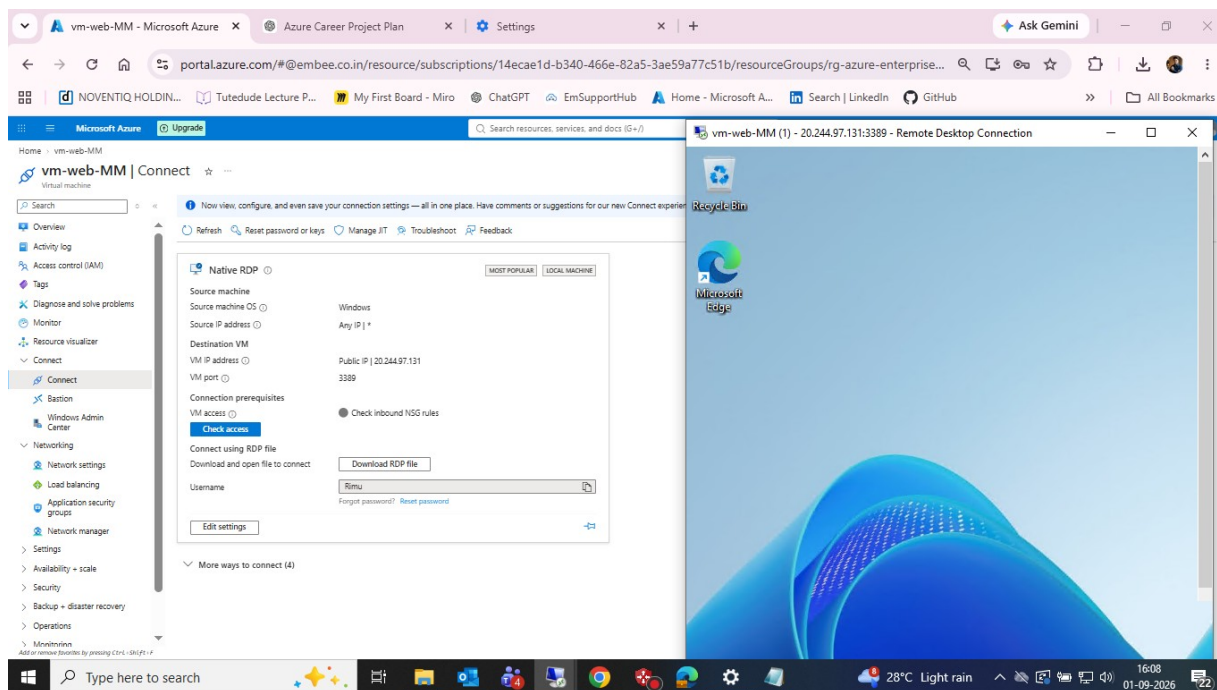
ESTIMATED STORAGE COST / MONTH 0.00 USD

COMPUTE COST / VCORE SECOND 0.000000 USD

1 There will be no charges for usage within the free limits. The database will be provisioned automatically when the free limits are reached.
2 Sometimes databases are billed in vCore seconds based on a combination of CPU and memory utilization. [Learn more about serverless billing.](#)

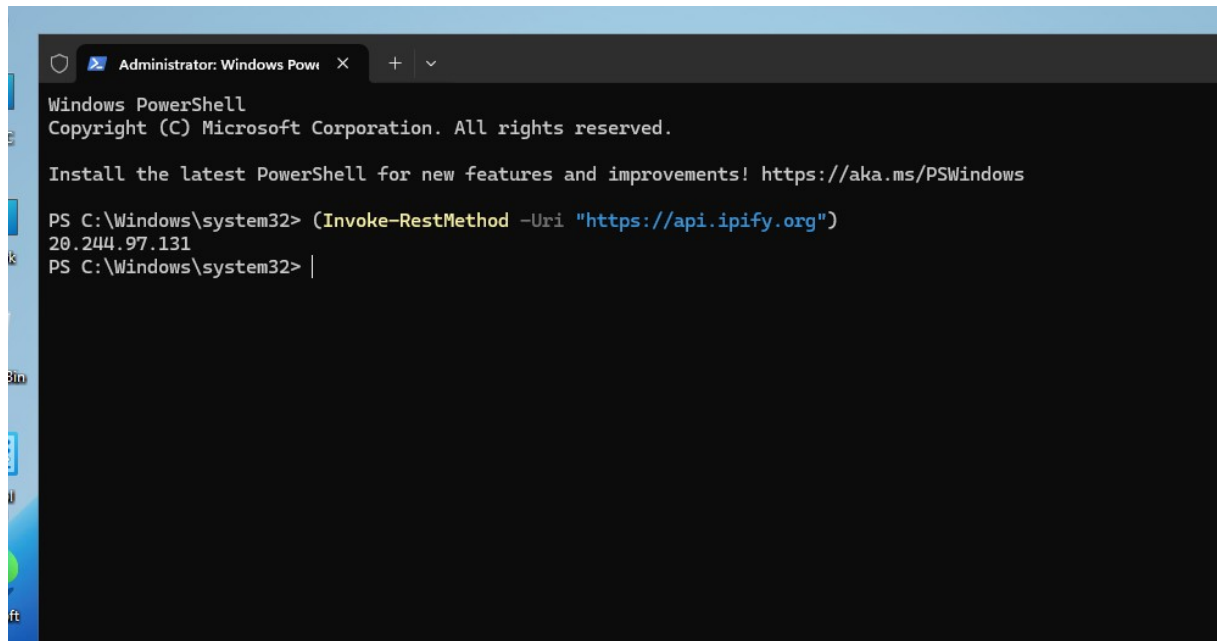
SQL Free offer applied with zero displayed compute/storage cost

Figure 10 — Successful RDP session to vm-web-MM



Successful RDP session to vm-web-MM

Figure 11 — Outbound public IP verified from Windows VM



```
Administrator: Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Windows\system32> (Invoke-RestMethod -Uri "https://api.ipify.org")
20.244.97.131
PS C:\Windows\system32> |
```

Outbound public IP verified from Windows VM

Figure 12 — VM Overview after deployment

The screenshot displays the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information. The main content area is titled 'vm-web-MM' and shows the 'Overview' tab. The overview includes a search bar, a list of actions (Connect, Start, Restart, Stop, Hibernate, Capture, Delete, Refresh, Scale, Open in mobile, Feedback), and a list of properties (Status, Location, Subscription, Subscription ID, Availability zone, Tags, Application Owner, Properties, Monitoring, Capabilities, Recommendations, Tutorials). The 'Properties' section shows details for the virtual machine, including the operating system (Windows (Windows Server 2025 Datacenter)), size (Standard B2ts v2 (2 vcpus, 1 GiB memory)), primary NIC public IP (20.244.97.131), virtual network/subnet (vm-web-MM/Network), DNS name (Not configured), health state (Ready), and time created (9/1/2026, 10:12 AM UTC). The 'Networking' section shows the public IP address (20.244.97.131) and the associated public IP (vm-web-mm993). The 'Notifications' panel on the right shows a 'Deployment succeeded' message and a 'Credit remaining' notification.

vm-web-MM - M x Pricing Calculator x MizanurMollah x Azure Career Pro x New Tab x New Tab x Ask Gemini x

portal.azure.com/#@embee.co.in/resource/subscriptions/14ecae1d-b340-466e-82a5-3ae59a77c51b/resourceGroups/rg-azure-enterprise_MM/...

Microsoft Azure Upgrade Search resources, services, and docs (G+/)

Home > Compute infrastructure > Virtual machines

vm-web-MM Virtual machine

Help me copy this VM in any region Manage this VM with Azure CLI

Search

Connect Start Restart Stop Hibernate Capture Delete Refresh Scale Open in mobile Feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Monitor

Resource visualizer

Connect

Networking

Settings

Availability + scale

Security

Backup + disaster recovery

Operations

Monitoring

Automation

Help

rg-azure-enterprise_MM

Windows (Windows Server 2025 Datacenter)

Status: Running

Location: Central India (Zone 2)

Subscription: (move) My Test Azure subscription 1

Subscription ID: 14ecae1d-b340-466e-82a5-3ae59a77c51b

Availability zone: 2

Tags (edit)

Application Owner: MM

Properties Monitoring Capabilities (0) Recommendations Tutorials

Virtual machine

Computer name: vm-web-MM

Operating system: Windows (Windows Server 2025 Datacenter)

VM generation: V2

VM architecture: x64

Agent status: Ready

Agent version: 2.7.41621.1216

Networking

Public IP address: 20.244.97.131 (Network interface: vm-web-mm993)

1 associated public IPs

Public IP address (IPv6): -

Private IP address: 10.0.1.4

Private IP address (IPv6): -

Notifications

More events in the activity log > Dismiss all

Deployment succeeded

Deployment "CreateVirtualMachine-vm-web-MM" to resource group "rg-azure-enterprise_MM" was successful.

Go to resource Go to resource group

a few seconds ago

₹19,109.25 credit remaining

Subscription "My Test Azure subscription 1" has a remaining credit of ₹19,109.25.

Upgrade to a Pay-As-You-Go subscription.

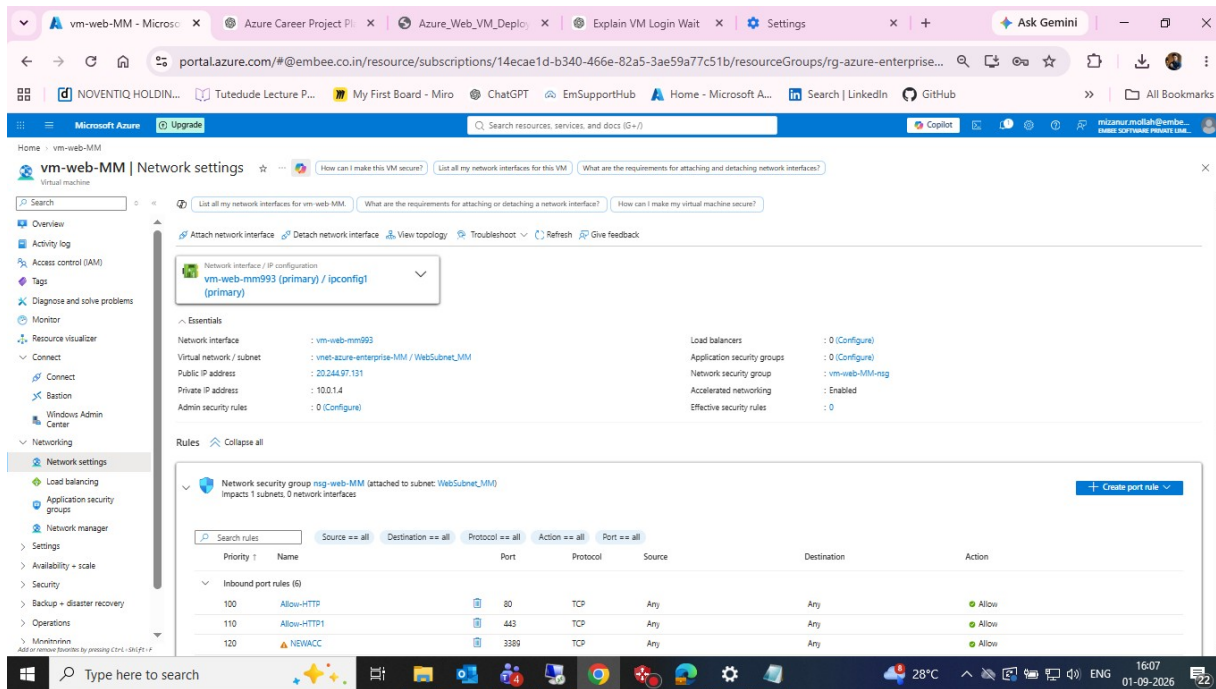
32 minutes ago

Type here to search

28°C 15:44 01-09-2026

VM Overview after deployment

Figure 13 — VM Network Settings



VM Network Settings

21. Next Step

Continue inside vm-web-MM with the Microsoft SQL Server Developer Edition installation. Capture each installer stage and the final SSMS/T-SQL evidence. After those screenshots are available, this document can be upgraded from a combined Azure SQL + planned SQL Server-on-VM record into a fully verified SQL Server-on-VM implementation document.